

Umang Singh

Software engineer with **2+ years at Oracle** building cloud systems, **multiple hackathon wins**, and 3 peer-reviewed papers.

[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | umang2singh@gmail.com | +91 9561537119 | India

TECHNICAL SKILLS

Languages & Backend: Java, Python, Go, SQL, Kotlin, Dart, Spring Boot, FastAPI, REST APIs, asyncio, Postgres, Flutter

Cloud Systems: GCP (Vertex AI, GKE, Cloud Run), AWS (Bedrock, Lambda, S3), Oracle Cloud Infrastructure

DevOps & Observability: Docker, Kubernetes, Helm, Terraform, Jenkins, GitHub Actions, CI/CD, distributed tracing

AI/ML: Agentic AI (LangGraph), MCP, RAG (pgvector), fine-tuning, PyTorch, Hugging Face, guardrails, evaluation harnesses

WORK EXPERIENCE

Oracle

Mumbai, India

Software Engineer

Jul 2024 – Present

- Built and own **4 components** of an anti-money-laundering SaaS: Java microservices for provisioning and metering, database migrations and a **FastAPI** service. Migrated metering from Spring Boot to Helidon, cutting startup **~40%** and memory **~25%**.
- Improved the reliability of an ML batch job so it recovers on its own from **database outages of up to 15 minutes** and **Kubernetes node replacement**, using database-request replay and idempotent restarts.
- Achieved easier maintenance and testing by splitting a **30,000+ line Python monolith (200+ APIs)** into separate analytics and data-access layers in about **3 weeks** instead of an estimated **2 months**, using AI-assisted development.

Project Intern

Jun 2023 – Jul 2023

- Built an automated data-integration pipeline with **150+ field mappings**, bringing customer, account and transaction data together into a unified case-management system for financial-crime investigations.

Futurepreneur Tech

Mumbai, India

Technical Lead

Feb 2023 – May 2023

- Shipped a hostel-discovery app on **Flutter** and **Firebase** with optimistic UI and onboarded **20+ hostels**.
- Secured **INR 100,000** in funding from the college principal and led a **15+ person team** across UI/UX, frontend and backend.

ACHIEVEMENTS

- 1st Runner-Up** at the [Google Agentic AI Hackathon 2025](#) (57,000+ registrants) and the [Japan-India Hackathon 2024](#).
- Winner** of [Smart India Hackathon \(SIH\) 2023](#), run by the Government of India with **250,000+** participants.
- Finalist** at national hackathons: [BIAL Genie](#) by **Microsoft** (top 8 of **11,000+**) and [ONDC-Google Build for Bharat](#).
- Ranked **1,081 of 24,578 (top 4.4%)** in LeetCode [Weekly Contest 352](#), with **700+** problems solved across 500+ active days.

PROJECTS

[ChronoTrace](#): AI Agent for Flaky-Test Repair | Python, asyncio, Amazon Bedrock, Strands Agents

Aug 2026 – Sep 2026

- Built a **flaky-test AI agent** optimized for race conditions in concurrent code that finds the steps running out of order and fixes their ordering, while the same LLM alone masked the race with a sleep or retry in **86%** of cases

[KrishiBodha](#): Agentic AI Assistant for Farmers | Gemini, Gemma, Vertex AI, Google ADK, Flutter

Jun 2025 – Aug 2025

- Led the team** behind a multilingual voice assistant for farmers covering crop disease, market alerts, subsidies, and crop and soil health from **Sentinel-2** satellite imagery, running **11 AI agents** on Gemini with an **on-device Gemma** model for offline use

[VidyuRakshak](#): Power-Line Safety from Satellite Imagery | Google Earth Engine, Random Forest, AWS

Dec 2023 – Aug 2024

- Trained a **Random Forest** model that estimates tree height from 7 vegetation indices in satellite imagery so power utilities can spot trees growing too close to their lines, validated against elevation data on **107,000** records
- After it won SIH 2023, [Power Grid Corporation of India](#) signed an agreement to develop it further, and the team kept the IP

PUBLICATIONS

- "[Advanced Deep Learning Techniques for Early Detection of Oral Cancer: A Multimodal Approach](#)"; ISAI, Springer. Mar 2025
- "[Exploring Challenges and Results in Crop Type Mapping Using Satellite Imagery](#)"; IEEE InGARSS. First author. Dec 2024
- "[Detecting Tree Height and Addressing Challenges Along Power Line Corridors...](#)"; ICICBDA, Springer. Jun 2024

LEADERSHIP AND EXTRACURRICULARS

- Mentored 15 AI/ML teams** at [HackSync 2.0](#), TSEC's hackathon, on problem framing and model choice.
- Conducted a database workshop** for **100+ students** at a TSEC [alumni session](#) on connecting Java, Python and MERN apps.
- Taught mathematics** to underprivileged students for 2 years with [U&I NGO](#)'s Project Sunlight.

EDUCATION

University of Mumbai | [College Hall of Fame Award](#) (30 of 600+ students)

Mumbai, India

Bachelor of Engineering, Computer Engineering | **GPA: 8.76/10**

2020 – 2024